



Nodes in Kubernetes

Introduction

A **Node** in Kubernetes is a **physical or virtual machine** that runs application workloads.

- What are Nodes in Kubernetes?
- What is the role of Nodes in a Kubernetes cluster?
- How do Nodes work together to run applications?

This guide will help you **understand Kubernetes Nodes**, their components, and how they function within a cluster.

Section 1: Learn

What is a Node?

A **Node** is a **worker machine** that runs **containers inside Pods**. Kubernetes manages multiple **Nodes** to ensure **high availability and scalability**.

Feature	Description
Computing Power	Provides CPU and memory for running applications
Container Execution	Runs Docker, containerd, or other runtimes
Networking	Connects Pods with services
Storage Management	Mounts persistent storage for applications
Fault Tolerance	Kubernetes shifts workloads when a Node fails



Types of Nodes

- **Master Node** – Controls and manages the cluster.
- **Worker Node** – Executes application workloads.

Each **Kubernetes cluster** has **one or more Master Nodes** and **multiple Worker Nodes**.

Node Components

A Worker Node contains **three key components**:

Component	Function
Kubelet	Ensures Pods are running as expected
Kube Proxy	Manages networking and load balancing
Container Runtime	Runs containers (Docker, containerd)

Interesting Fact

Google originally designed Kubernetes to run on **thousands of Nodes** across **multiple data centers**.

Section 2: Practice

1. Checking Available Nodes

To view all Nodes in a cluster, run:

```
kubectl get nodes
```

To get detailed information about a specific Node:



```
kubectl describe node <node-name>
```

2. Simulating a Node Failure

To **cordons** a **Node** (prevent scheduling new Pods on it):

```
kubectl cordon <node-name>
```

To **drain** a **Node** (move workloads to other Nodes):

```
kubectl drain <node-name> --ignore-daemonsets --delete-emptydir-data
```

To bring a Node back online:

```
kubectl uncordon <node-name>
```

3. Viewing Node Resource Usage

To check Node resource consumption:

```
kubectl top nodes
```

Section 3: Know More

Frequently Asked Questions

1. **What is the difference between a Pod and a Node?**
 - A **Pod** runs one or more containers, while a **Node** runs multiple Pods.
2. **Can a Node run multiple Pods?**
 - Yes, a single Node can host **multiple Pods**, depending on its resource limits.
3. **What happens if a Node fails?**



- Kubernetes detects failure and **automatically reschedules Pods** to healthy Nodes.
4. **How does Kubernetes add new Nodes?**
- Kubernetes can **automatically scale** by adding Nodes dynamically.
5. **Do all Nodes need to be identical?**
- No, Kubernetes allows Nodes with **different CPU, memory, and storage configurations**.
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Conclusion

Nodes are the **building blocks** of a Kubernetes cluster. They provide **compute power, networking, and storage** to run applications.

Understanding **how Nodes function** helps in managing **Kubernetes clusters efficiently**.